



Urgency-Aware QoE-Driven Cooperative 3D Trajectory Planning for UAV Fleets in Post-Disaster Communications via Multi-Agent DRL

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Outline

1. Introduction
2. Research Proposal
3. Results
4. Conclusion

UAVs: Transforming Industries, Expanding Human Capabilities

What are UAVs?

Unmanned Aerial Vehicles (UAVs), commonly known as drones, are remotely controlled or autonomous aircraft.

Why UAVs Matter?

UAVs offer unique capabilities to perform tasks with enhanced speed, safety, and efficiency, especially in hazardous, remote, or data-intensive environments.

Their Diverse Applications:

- Agriculture: Crop monitoring, spraying, and yield assessment
- Logistics: Fast and contactless delivery
- Surveillance: Border security, traffic monitoring, and disaster response
- Filmmaking: Aerial cinematography and photography
- Infrastructure: Inspection of bridges, towers, and power lines
- Environmental Monitoring: Wildlife tracking and air quality assessment



Motivation: Enhancing Emergency Connectivity with UAVs

Infrastructure Breakdown: The Critical Problem

- **Network Collapse:** During natural or man-made disasters, terrestrial communication infrastructure is often damaged or completely destroyed.
- **Urgency of Real-Time Data:** Immediate, reliable connectivity is a strict prerequisite for situational awareness and organizing rescue operations in high-stakes environments.



UAV-Assisted Communication: The Swarm Solution

- **Rapid Deployment:** Unmanned Aerial Vehicles (UAVs) function as agile aerial base stations (ABSs) that instantly restore communication [1].
- **Altitude Advantage:** ideal for covering hard-to-reach areas. Overcoming urban terrain and physical obstacles through altitude adjustment.

Motivation: Cooperative 3D Trajectory Design

The Challenges:

- **Lack of Urgency-Aware Prioritization:** Conventional networks utilize homogeneous service provisioning. In disaster scenarios, treating critical emergencies and standard traffic with equal priority induces severe service latency for vulnerable users.
- **Discrepancy Between QoS and QoE:** Optimizing for raw network throughput (Quality of Service) fails to accurately capture the subjective communication satisfaction (Quality of Experience) required during crises.
- **Inflexibility of Static Deployments:** Fixed terrestrial infrastructure and pre-planned, static UAV trajectories suffer from severe coverage blind spots and cannot adapt to unpredictable 3D obstacles or dynamic user mobility.

The Solution: Cooperative 3D Trajectory Design

- **Autonomous 3D Spatial Adaptation:** A MARL-driven UAV fleet dynamically optimizes its 3D coordinates (X, Y, Altitude) to proactively bypass obstacles, adapt antenna footprints, and bridge coverage gaps in real-time.
- **Priority-Driven Dynamic Coverage:** Flight trajectories are continuously optimized to target high-urgency user clusters, maximizing the global network QoE while minimizing service latency for mission-critical tasks.

Quality of Experience Model: Mean Opinion Score (MOS)

Objective: Maximize user satisfaction via demand-centric resource management [8].

1. Logarithmic Mapping (Continuous score x): The user data rate $R_{i,t}$ is mapped to a continuous quality score $x_{i,t}$

$$x_{i,t} = a \cdot \log_{10} \left(\frac{R_{i,t}}{b} \right)$$

- Range: $\theta_1 = 0.5$ Mbps (MOS 1) to $\theta_2 = 5.0$ Mbps (Saturation, MOS 5).

where $a = \frac{x_2 - x_1}{\log_{10}(\theta_2/\theta_1)}$ $b = \frac{\theta_1}{10^{x_1/a}}$

2. Discrete Mapping to MOS: MOS is defined on a 1–5 scale to reflect human-perceived quality:

$$MOS_{n,t} = \begin{cases} 0, & R_{i,t} \leq R_{th} \\ 1, & R_{i,t} > R_{th} \text{ and } x_{i,t} < 1.5 \\ 2, & 1.5 \leq x_{i,t} < 2.5 \\ 3, & 2.5 \leq x_{i,t} < 3.5 \\ 4, & 3.5 \leq x_{i,t} < 4.5 \\ 5, & x_{i,t} \geq 4.5 \end{cases}$$

- Outage (MOS = 0): $R_{i,t} \leq 100$ kbps (Hardware limit).
- Discrete Scale: 1 (Poor) to 5 (Excellent) based on $x_{i,t}$ thresholds.

Proposed Priority-Aware Model: Embedding Urgency into Perceived MOS

The Relief Effect in Disasters

- **The Limitation of Standard MOS:** Traditional MOS assumes satisfaction requires high bandwidth. In a crisis, establishing even a basic, low-bandwidth connection provides immense psychological and life-saving relief to emergency users.
- **The Goal:** We must mathematically trick the RL agents into valuing this "basic relief connection" as highly as a perfect connection for a normal user.

Proposed Priority Factor $\alpha_{i,t}$

- **Mathematical Definition:** $S_{i,t} = \min(\alpha_{i,t} \text{MOS}_{i,t}, S_{\max})$
- **Configurable Architecture:** α is a flexible policy parameter. While we use representative values to design the trajectory framework, it is built to seamlessly ingest exact empirical values from future sociological disaster surveys.

Normal Users ($\alpha = 1.0$)	Important Users ($\alpha = 1.5$)	Emergency Users ($\alpha = 2.0$)
Status: Baseline civilian traffic. Trajectory Impact: The baseline. The UAV must fly directly overhead and provide a perfect raw signal ($\text{MOS} = 5.0$) to reach maximum satisfaction.	Status: First responders / Logistics. Trajectory Impact: Reaches saturation early. The UAV can achieve max satisfaction with only a good signal ($\text{MOS} \approx 3.3$), freeing the UAV to hover higher and cover a wider footprint.	Status: SOS beacons / critical victims. Trajectory Impact: Models the "Relief Effect." Even a distant signal ($\text{MOS} = 2.5$) is elevated to max satisfaction, guaranteeing the UAV prioritizes immediate baseline connectivity from afar.

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Literature Review & The Research Gap

1. Network-Centric & Uniform QoE Optimization

- **Approach:** Maximizing aggregate metrics like throughput, sum-rate, or basic Mean Opinion Score (MOS) [2-8].
- **Limitation:** Treats all users uniformly, ignoring the heterogeneous, life-threatening urgency of disaster zones.

2. Priority-Aware Policies (& Author's Previous Work)

- **Early Approaches:** Relied on rigid, discrete labeling (e.g., binary IoT classifications) without continuous spatial adaptability [9,10].
- **Author's Previous Work:** Solved this rigidity by introducing a continuous Dynamic Priority Scaling Factor (α) and Multi-Channel Heatmaps trained via PPO.
- **Remaining Challenges:** Limited to a Single-Agent in a 2D, obstacle-free space. It cannot physically scale to multi-wave disaster zones, and 2D models ignore real-world signal blockage.

3. The 3D Multi-Agent Challenge (Current Work)

- **The Goal:** Transition to a cooperative Multi-Agent fleet operating in a physically realistic 3D Urban Environment (with buildings, foliage, and Non-Line-of-Sight (NLoS) shadowing).
- **The Barrier:** Standard MARL suffers from severe Sparse Rewards in massive 3D environment and the Credit Assignment (Lazy Agent) problem when evaluating teamwork.



The Research Question: How do we develop a cooperative multi-agent trajectory framework that enhanced fleet-wide 3D coverage while rapidly responding to emergency localization and services in a large-scale post-disaster scenario?

Research Objectives

Overarching Goal: To propose and evaluate a two-stage hybrid control framework (**K-Means + MAPPO**) to optimize resilient, priority-aware communication links via a UAV swarm in urban post-disaster networks.

1. **Hybrid Deployment Strategy:** Combine K-Means clustering (macro-allocation) with MARL (micro-trajectory) to eliminate cold-start delays and accelerate response times.
2. **Heterogeneous QoE Optimization:** Maximize Perceived MOS across diverse urgency tiers using dynamic UAV trajectories, altitude control and 3D adaptive beamforming.
3. **Enforce Mission-Critical Service Level Agreements (SLAs):** Minimize emergency response latency and enforce mandatory minimum-service durations for life-saving communication links.
4. **Robust 3D Spatial Navigation:** Train the swarm to autonomously bypass obstacles and severe Non-Line-of-Sight (NLoS) signal attenuation using a high-fidelity 3D ray-tracing model
5. **Comprehensive Benchmarking:** Validate swarm scalability and coverage efficiency through rigorous ablation studies (vs. Single-Agent, Pure MAPPO, and Static Centroid).

System Model: 3D Urban Disaster Environment

Overview: In the absence of terrestrial infrastructure, a fleet of $M = 4$ UAVs operates as dynamic Aerial Base Stations, cooperating in 3D space to deliver priority-aware network coverage.

1. UAV Mechanics & Action Space:

- 3D Flight Control: 7 discrete actions (North, South, East, West, Hover, Ascend, Descend).
- Adaptive Beamforming: Altitude dynamics are physically coupled with antenna beamwidth and directional gain [11].

2. Communication & Environment Physics:

- A2G Channel: Air-to-Ground propagation featuring dynamic Rician fading [5].
- Obstruction Modeling: 3D ray-tracing with Non-Line-of-Sight (NLoS) building shadowing and indoor penetration loss [12].

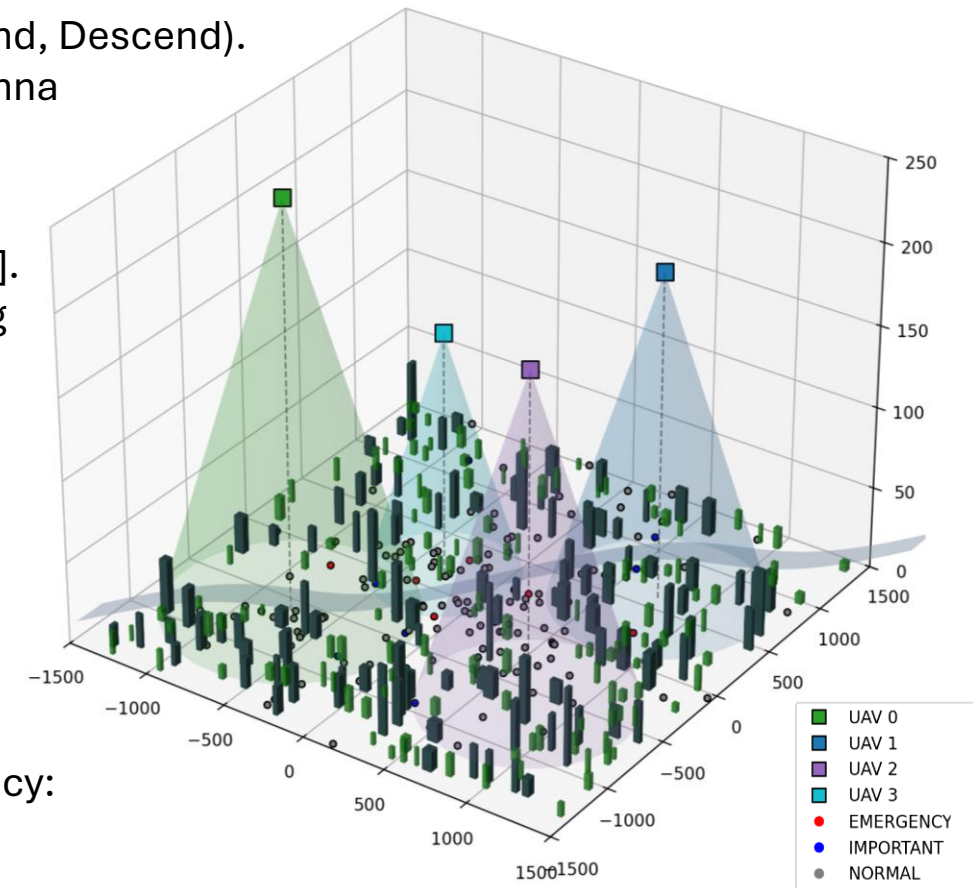
3. User Mobility & Priority Hierarchy:

- Mobility: Ground users follow a continuous Gaussian random walk.
- Priority Multiplier ($\alpha_{i,t}$): Users are dynamically classified into three tiers:
- Normal: 1.0 | Important: 1.5 | Emergency: 2.0

4. Priority-Aware QoE Model:

- We extend standard capacity models into a Perceived Quality of Experience (QoE) metric, scaling the raw Mean Opinion Score (MOS) by the user's urgency:

$$S_{i,t} = \min(\alpha_{i,t} \text{MOS}_{i,t}, S_{\max})$$



[11] 3GPP, "Study on Enhanced LTE Support for Aerial Vehicles," 3rd Generation Partnership Project (3GPP), Technical Report (TR) 36.777, Release 15, Dec. 2017.

[5] L. T. Hoang, C. T. Nguyen, H. D. Le, and A. T. Pham, "Multi-Agent Reinforcement Learning for Cooperative Trajectory Design of UAVBS Fleets in Terrestrial/Non-Terrestrial Integrated Networks," in *IEICE Communications Express*, vol. 13, no. 8, pp. 327-330, August 2024.

[12] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project (3GPP), Technical Report (TR) 38.901, Release 16, Dec. 2019.

Problem Formulation: QoE Maximization

1. Objective Function: Optimize 3D trajectory of the UAV fleets to maximize cumulative priority-aware user satisfaction (QoE) over the mission duration T .

$$\mathcal{P}: \max_{\mathbf{q}} \sum_{t=1}^T \sum_{i=1}^N S_{i,t}$$

where Perceived MOS is by the time-varying priority weight $\alpha_{i,t}$:

$$S_{i,t} = \min(\alpha_{i,t} \text{MOS}_{i,t}, S_{\max})$$

2. 3D Kinematics & Beamforming Constraints: UAVs are bounded by maximum velocity, 3D urban limits, and altitude-coupled antenna beamwidths (θ).

- Mobility: $\|\mathbf{q}_{m,t+1} - \mathbf{q}_{m,t}\| \leq v_{\max} \Delta t, \forall m, t$
- Boundary: $\mathbf{q}_{m,t} \in [X_{\min}, X_{\max}] \times [Y_{\min}, Y_{\max}] \times [Z_{\min}, Z_{\max}], \forall m, t$
- Coupled Beamwidth: $\theta_{m,t} = f(z_{m,t})$, mapping $z \in [100m, 250m]$ to $\theta \in [60^\circ, 120^\circ]$

Problem Formulation (Con.): SLAs

3. Mission-Critical SLA Constraints:

Enforcement of strict stability windows for critical users. Let $c_{i,t} \in \{\text{Critical}, \text{Normal}\}$ represent the user state. Critical users must accumulate a mandatory minimum service duration $T_{\min}^{c_i}$ where $\text{MOS}_{i,t} \geq 1$ and $c_{i,t} = \text{Critical}$ before the incident is resolved.

Tracking Service Accumulation ($\tau_{i,t}$):

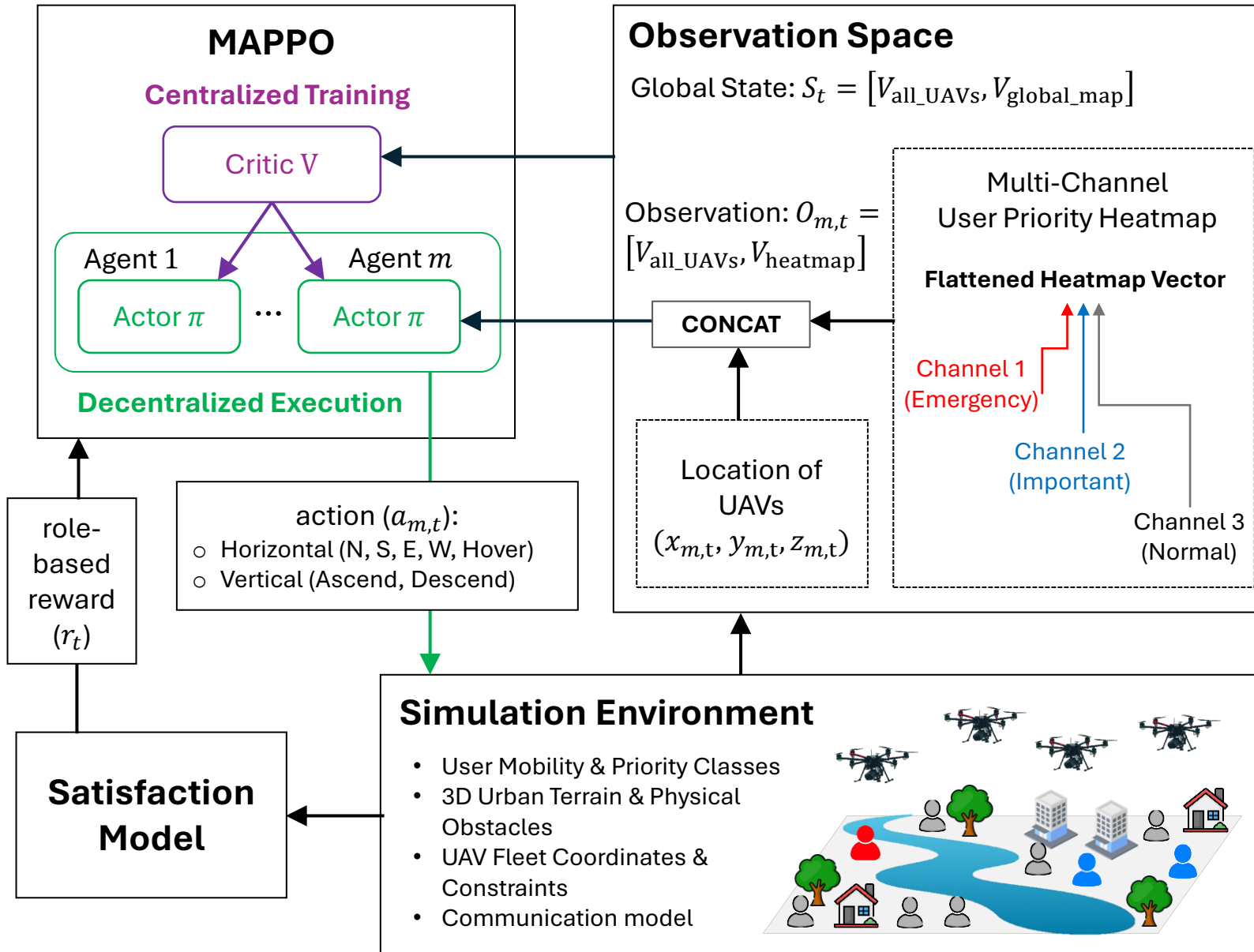
$$\tau_{i,t+1} = \begin{cases} \tau_{i,t} + 1, & \text{if } \text{MOS}_{i,t} \geq 1 \wedge c_{i,t} = \text{Critical} \\ \tau_{i,t}, & \text{otherwise} \end{cases}$$

User Severity State Transition ($c_{i,t+1}$):

$$c_{i,t+1} = \begin{cases} \text{Normal}, & \text{if } \tau_{i,t} \geq T_{\min}^{c_i} \\ c_{i,t}, & \text{otherwise} \end{cases}$$

- Hard SLA (Emergency): Failure to establish service results in a significant, time-dependent delay penalty.
- Soft SLA (Important): High reward priority but no latency penalty, allowing the swarm to dynamically balance coverage trade-offs.

MAPPO Architecture: CTDE Training & Feedback Loop



- **Framework:** Multi-Agent Proximal Policy Optimization (MAPPO)-based Centralized Training, Decentralized Execution (CTDE).
- **Key Metrics:** Jointly considers QoE-driven user satisfaction and multi-channel priorities.
- **Environment:** Tested in dynamic disaster scenarios with complex 3D urban landscapes and obstacles.
- **Objectives:** This continuous feedback loop allows the system to optimize the 3D trajectory of the UAV fleets in real-time. It learns to maximize priority-aware Quality of Experience (QoE) under the complex urban disaster landscapes.

Intelligent Design (1): Two-Stage Hybrid Trajectory Architecture

The Sparse Reward Dilemma: How do we deploy a UAV swarm across a massive 3km x 3km disaster zone rapidly, without suffering from the slow, trial-and-error cold-start of pure Reinforcement Learning?

Stage 1: Initial Deployment (Macro-Trajectory Planning)

- **Method:** K-Means Clustering (Heuristic Warm-Start).
- **Mechanism:** Partitions the global user distribution into $K = 4$ distinct geographic zones based on population density.
- **Action:** Bypasses empty airspace by calculating direct trajectory vectors and deploying the UAVs to these centroids at maximum velocity.

Stage 2: Service Phase (Micro-Trajectory Optimization)

- **Method:** Multi-Agent PPO-CTDE (Full RL Control).
- **Mechanism:** Upon reaching the centroid target deployment radius, the heuristic override permanently terminates.
- **Action:** The MARL agents take full trajectory control, performing continuous 3D spatial tuning to adapt beamwidths, avoid obstacles, and dynamically maximize QoE.

Intelligent Design (2): Two-Tier Disaster Response Strategy

The Spatial Dilemma: How does the fleet respond to a sudden, high-priority emergency without abandoning the background civilian network and causing a city-wide blackout?

Core Principle: Soft Guidance, Not Hardcoded Control

The K-Means algorithm does not hijack flight controls. It acts as a dynamic spatial scanner, feeding target coordinates into the reward function as a "soft pull" while the RL agents retain full control of the 3D trajectory.

Tier 1: Critical Chasers (Urgency Response)

- **Dynamic Spatial Scanning:** The environment isolates critical users and groups them into Z_c distinct disaster zones based on proximity.
- **Targeting:** The K-Means generates Z_c centroids. The RL Chasers receive a strong reward incentive ($\lambda_{\text{guide}}^{\text{chaser}}$) to track these epicenters, prioritizing high-speed rescue maneuvers.

Tier 2: General Providers (Crowd Control)

- **Coverage Maintenance:** The remaining Z_p UAVs (where $Z_p = M - Z_c$) are automatically designated as Providers.
- **Targeting:** The algorithm filters out the emergency and evaluates the Potential Gain of the remaining unserved normal crowd, guiding Providers to the densest civilian areas.

Unified Reward Architecture with Dynamic Role Assignment (CTDE)

Objective: To eliminate lazy agents and enforce load-balancing, the CTDE architecture calculates a unique, localized reward ($r_{t,m}$) for each individual UAV m , blending shared teamwork with strict individual accountability.

1. Unified Base QoE (50% Teamwork / 50% Accountability): Because the Global teamwork component does not increase when UAVs overlap on the same user, it mathematically forces the swarm to spread out.

$$R_{QoE,m} = \lambda_{QoE} \left[\underset{\text{Global Teamwork}}{0.5 \frac{\sum_{i \in \mathcal{N}} S_{i,t} \cdot w_i}{S_{\max} \sum_{i \in \mathcal{N}} w_i}} + \underset{\text{Local Accountability}}{0.5 \frac{\sum_{i \in \mathcal{N}_m} S_{i,t} \cdot w_i}{S_{\max} \sum_{i \in \mathcal{N}} w_i}} \right]$$

2. Dynamic Role Formulas: At each time step t , the K-Means algorithm assigns UAV m a role, applying asymmetric coefficients to the navigation and safety components:

Role 1: Critical Chaser (Urgency Response)

$$r_{t,m} = R_{QoE,m} + \lambda_{\text{guide}}^{\text{chaser}} (d_{m,t-1} - d_{m,t}) - \lambda_{\text{delay}} \sum_{i \in \mathcal{E}_m} p_i \frac{\min(\tau_{i,t}, \tau_{\max})}{\tau_{\max}} - (\lambda_{\text{overlap}} \Omega_{m,t} + \lambda_{\text{collision}} K_{m,t})$$

Role 2: General Provider (Crowd Control)

$$r_{t,m} = R_{QoE,m} + \lambda_{\text{guide}}^{\text{provider}} (d_{m,t-1} - d_{m,t}) - (\lambda_{\text{overlap}} \Omega_{m,t} + \lambda_{\text{collision}} K_{m,t})$$

$\mathcal{N}$: Set of all users in the network.

$\mathcal{N}_m$: Subset of users physically connected to UAV m .

$\mathcal{E}_m$: Unserved emergency users assigned to UAV m .

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Experimental Settings

Mission: 4 UAVs maximize priority-weighted QoE satisfaction for 200 mobile users across a 3×3 km post-disaster urban area, balancing broad coverage with rapid response to critical users under realistic 3D fading channel conditions.

ENVIRONMENT & UAV

Simulation area

3000 × 3000 m, bound [−1500, 1500]

UAV configuration

$M = 4$, $H = 100\text{--}250$ m, $v_{\max} = 25$ m/s

Antenna beamforming

60°–120°, coupled to altitude

CHANNEL MODEL

Carrier & bandwidth

5.8 GHz, 20 MHz

Transmit power

25 dBm (UAV downlink)

Noise floor

−90 dBm thermal noise

Propagation model

3D FSPL + NLoS penalty, PL exp 3.2,
Rician $k = 50$

USER TRAFFIC & TRAINING

User mobility

150 users, $v = 1.5$ m/s
Gaussian random walk, $\sigma = 650$ m

DRL training

K-means + MAPPO-based CTDE, 10,000
episodes, 800 steps/ep.

Disaster wave schedule

Initial Normal users only →
Important → Emergency → Mixed all

PRIORITY MIX & REWARD WEIGHTING

Normal — 90% of users

Priority factor α : 1.0
QoE reward weight w : 0.2

change → 1.0 (new)

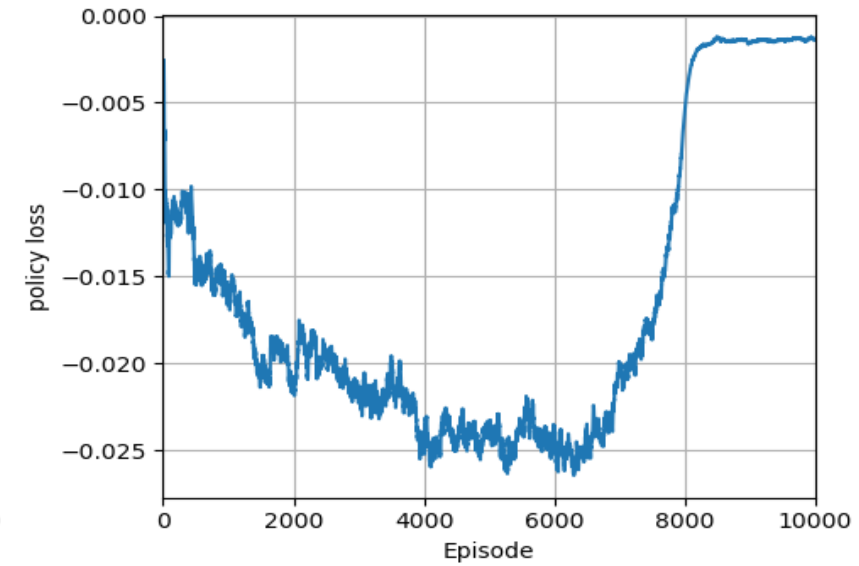
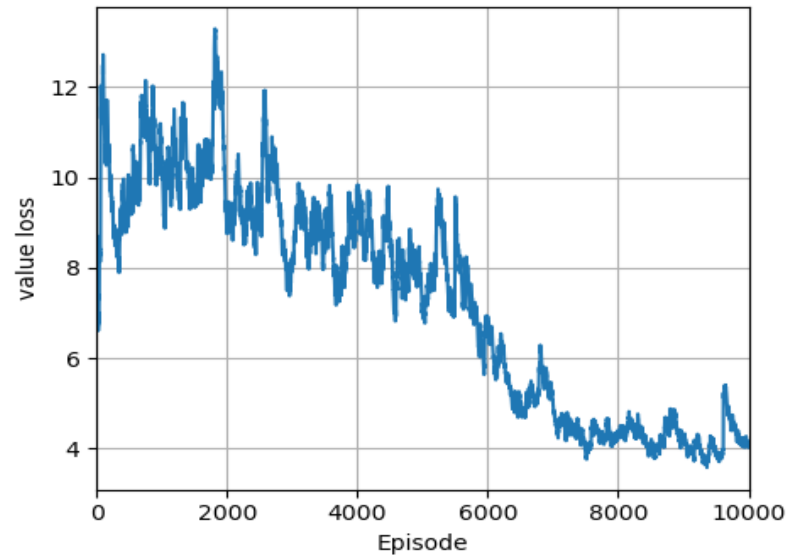
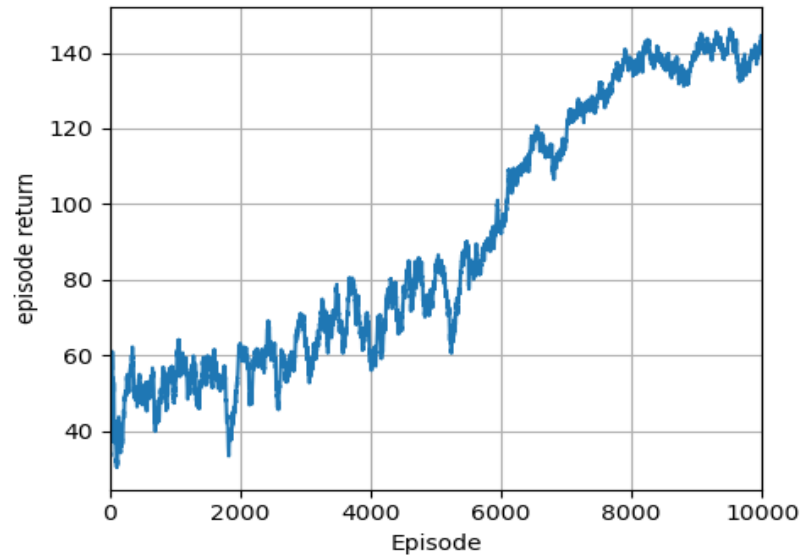
Important — 6% of users

Priority factor α : 1.5
QoE reward weight w : 6.0

Emergency — 4% of users

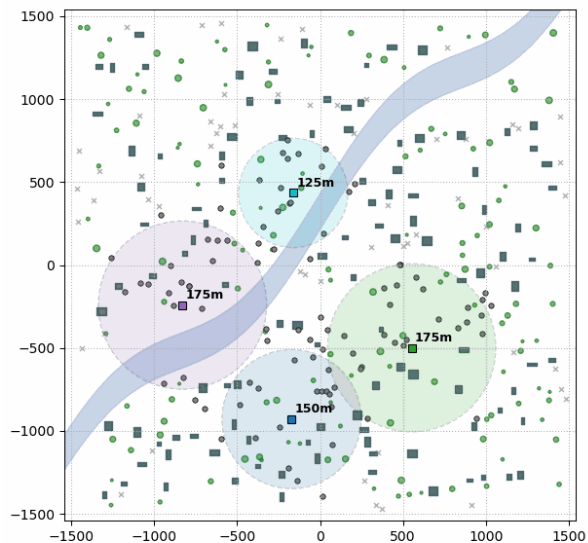
Priority factor α : 2.0
QoE reward weight w : 10.0

Results (1): Learning Performance

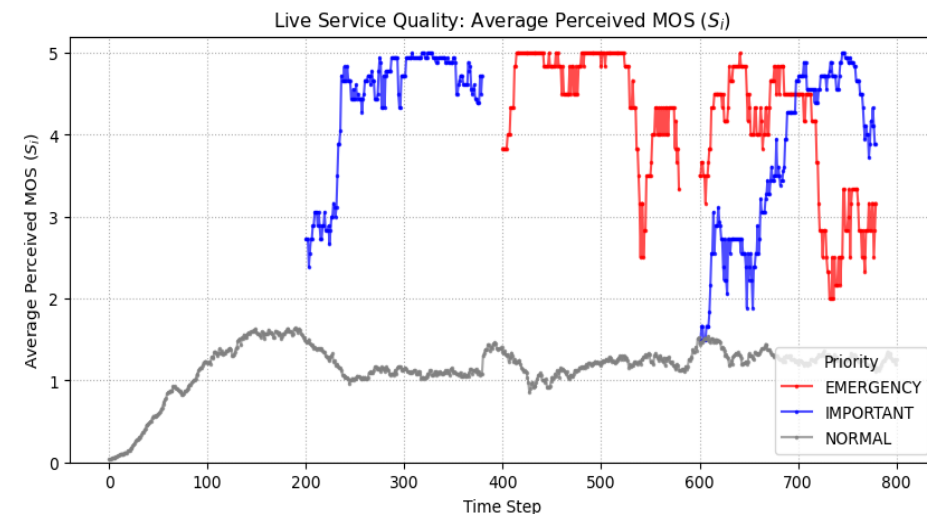
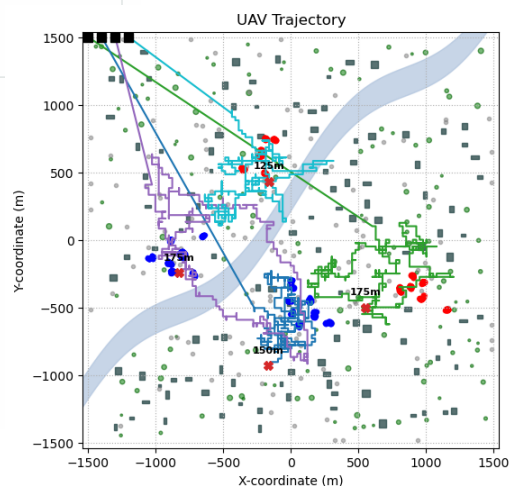


Action plan: Extend training to 20,000 episodes to ensure absolute policy convergence and lock down policy saturation.

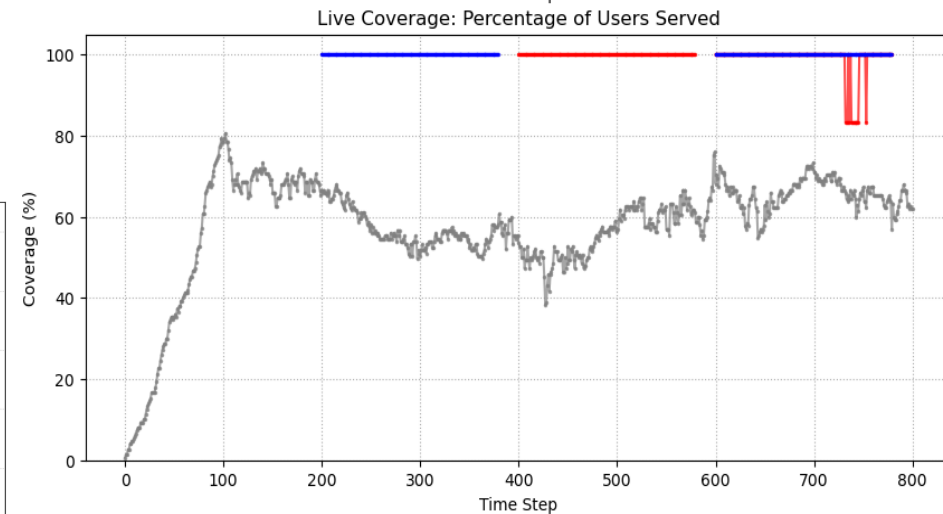
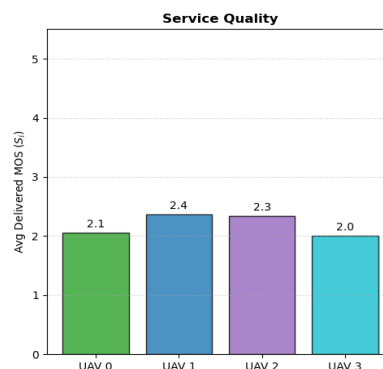
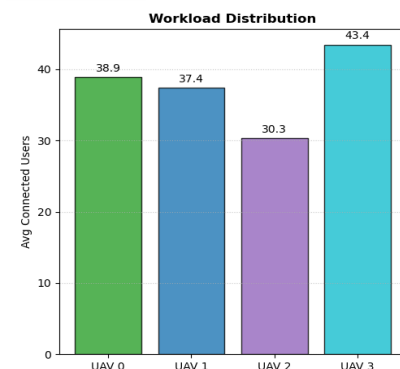
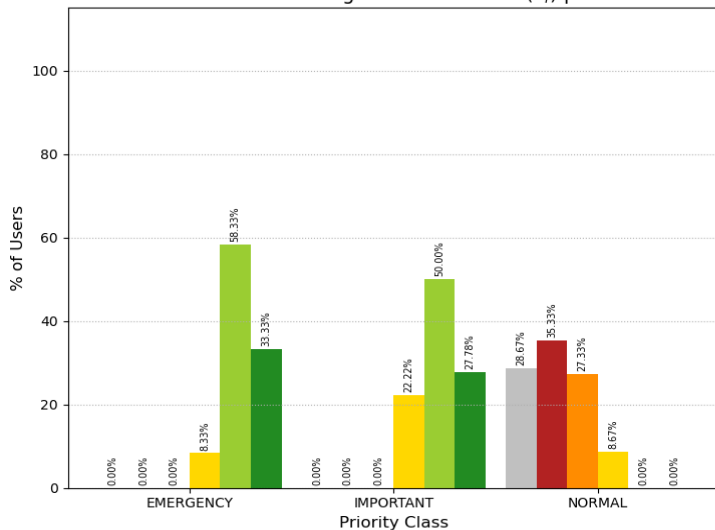
Results (2): Priority-Aware QoE Optimization



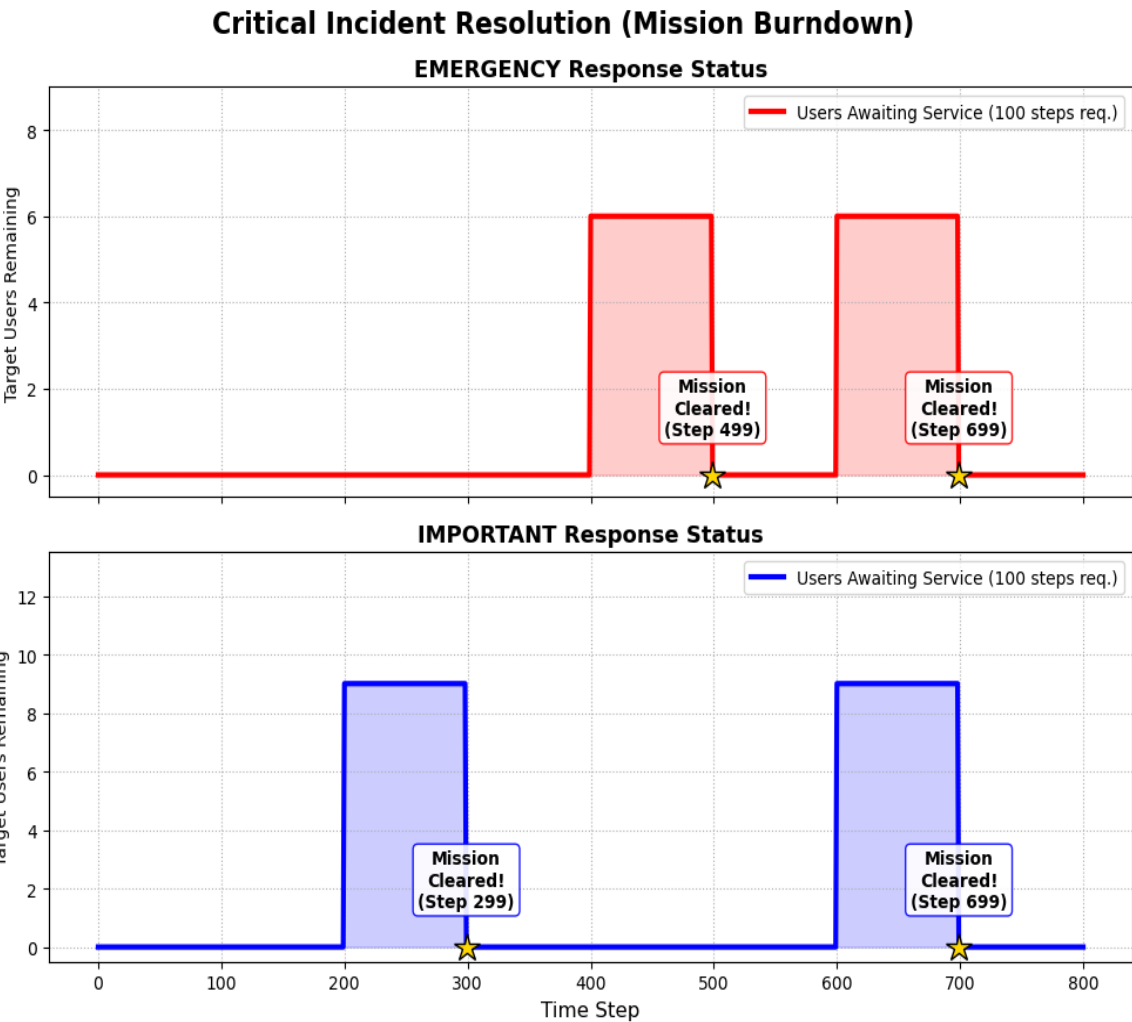
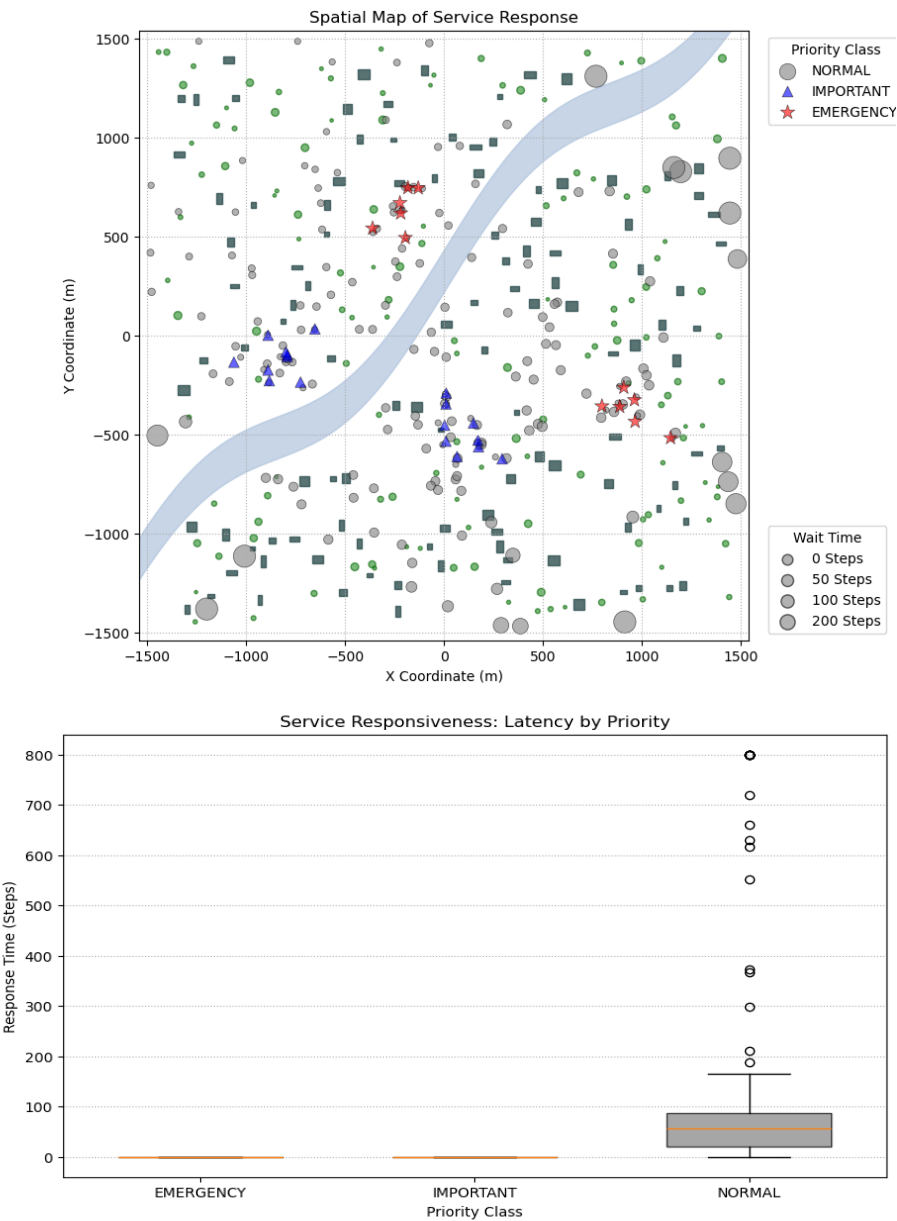
In this current setting, Normal $w = 0.2$ (too low) while Emergency $w: 10.0$
Agents don't do well to balance global coverage.



Distribution of Average Perceived MOS (S_i) per User

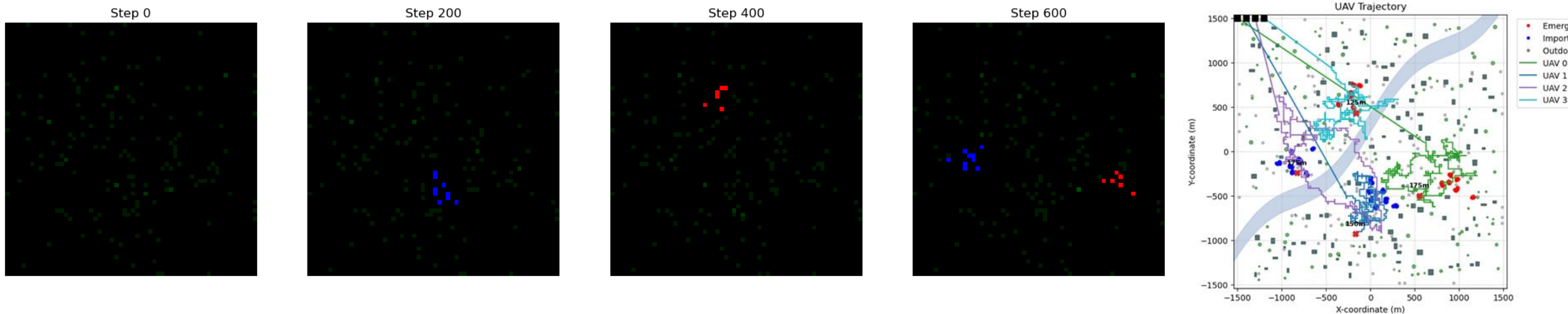


Results (3): Service Responsiveness

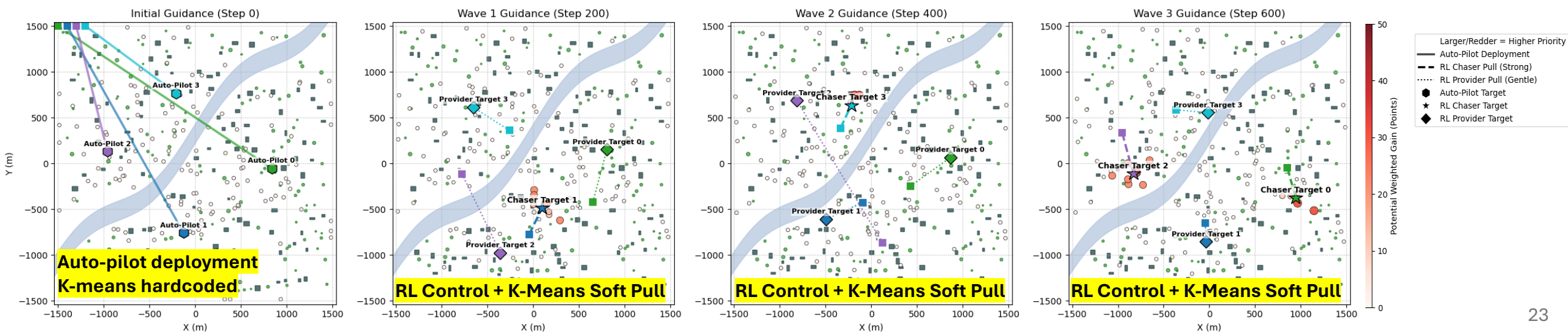


Results (4): Dynamic Spatial Tracking & Role Execution

User Heatmap (what agents see): All agents' shared observation

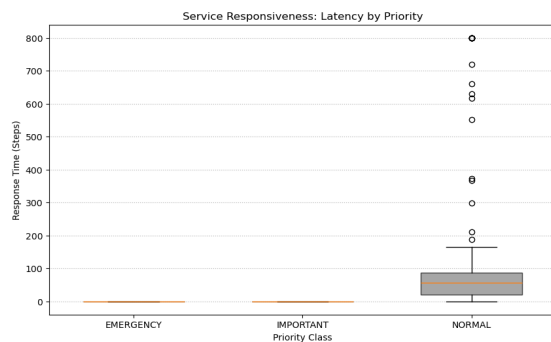
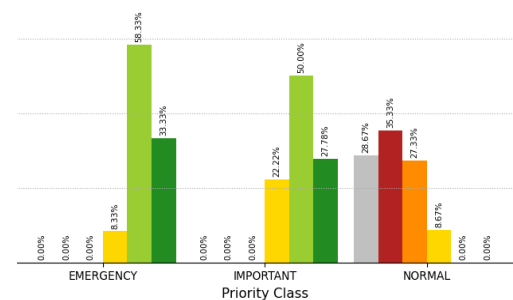
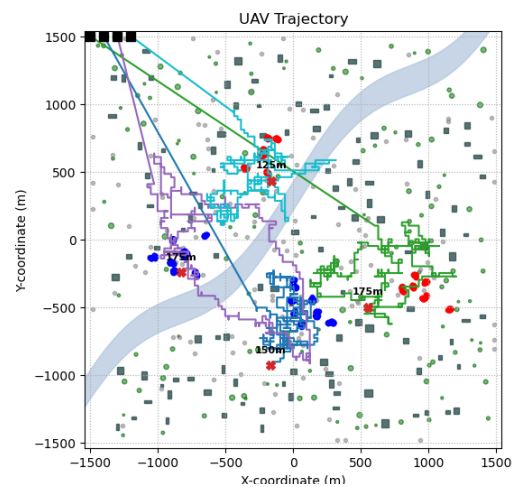


Potential Gain (K-means clustering soft guidance)

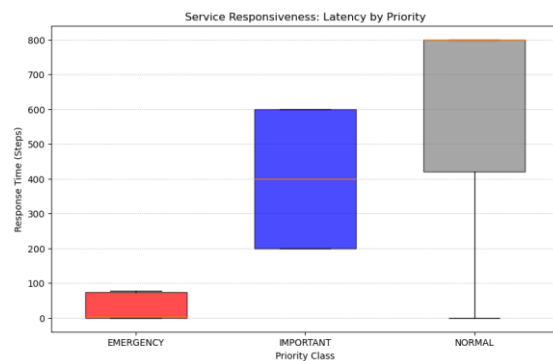
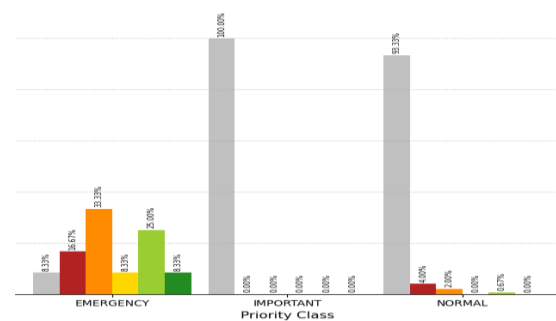
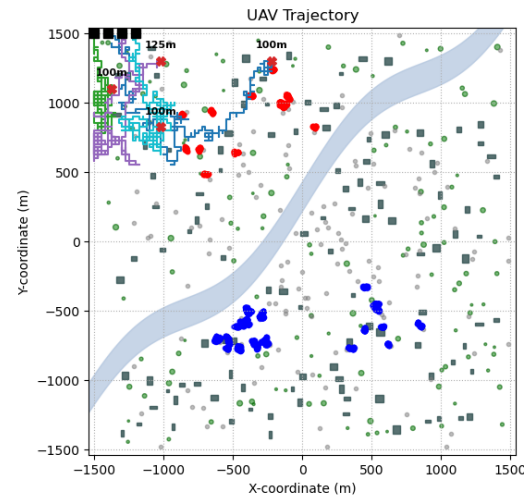


Results (5): Algorithmic Benchmarks

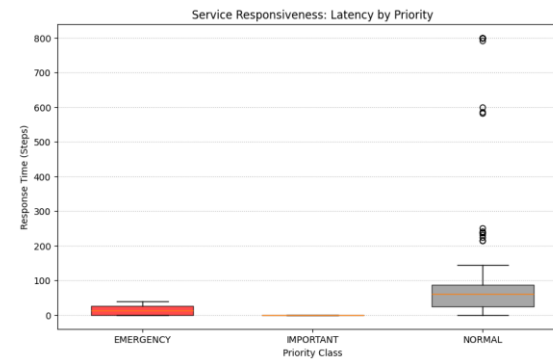
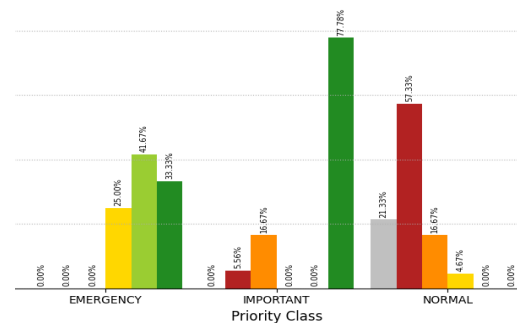
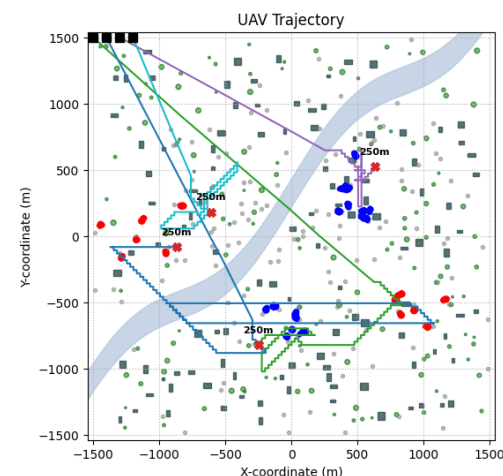
Proposed Framework



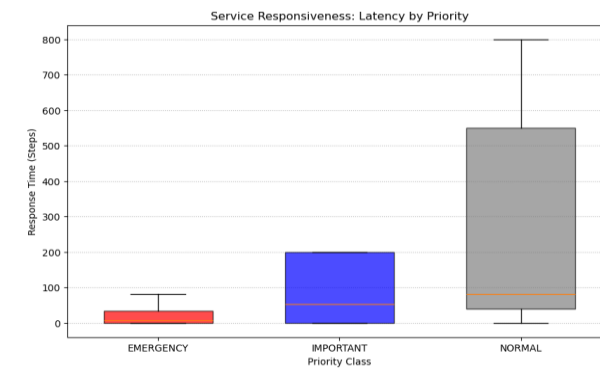
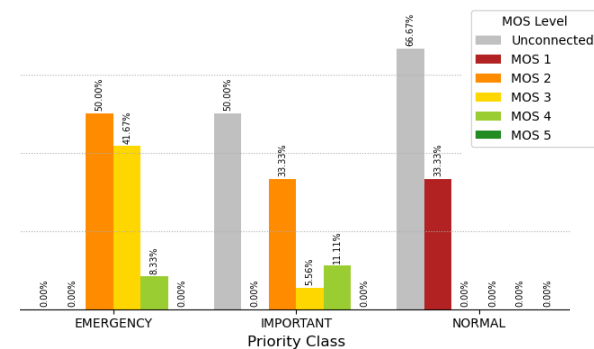
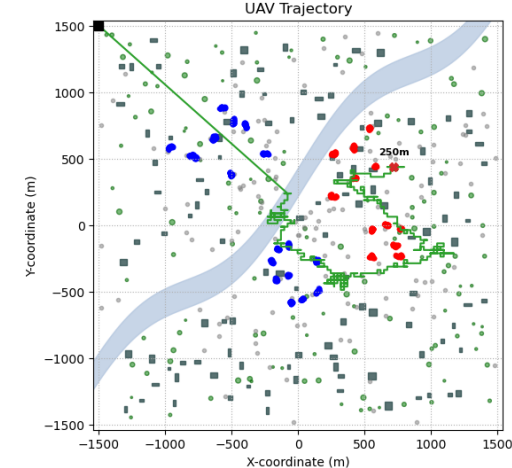
Pure MAPPO



Dynamic K-means



Single Agent



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Conclusion

1. **Hybrid Architecture Validated:** Successfully integrated K-Means + MAPPO to eliminate RL cold-start delays through macro-clustering and micro-trajectory optimization.
2. **Intelligent 3D Trajectory Adaptation:** The UAV swarm autonomously optimizes 3D flight paths, dynamic altitude, and adaptive beamforming to mitigate NLoS blockages and maximize priority-aware QoE.
3. **Strict SLA & Priority Enforcement:** The priority-aware framework actively targets critical users, successfully minimizing emergency response latency and guaranteeing mandatory minimum service durations.
4. **Benchmark Superiority:** Ablation studies confirm the proposed framework significantly outperforms Pure MAPPO, Single-Agent, and Static Baselines in both aggregate QoE and rapid emergency response.

Next Step:

- Fine-tune priority-tier reward weights to achieve a more robust balance in global network coverage.
- Enhance the baseline MAPPO framework with soft-center guidance to provide a more rigorous trajectory comparison.

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- [1] N. Zhao et al., "UAV-Assisted Emergency Networks in Disasters," in IEEE Wireless Communications, vol. 26, no. 1, pp. 45-51, February 2019, doi: 10.1109/MWC.2018.1800160.
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Thank You.
